

Department of Mechanical and Production Engineering

Aarhus University

# Varmetransmission

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| <b>Author</b>     | Noah Rahbek Bigum Hansen         | 202405538  |
| <b>Course</b>     | Varmetransmission                | 290221U002 |
| <b>Instructor</b> | Pourya Forooghi & Mugisa Y. Kano |            |
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## Lecture 1: Introduction to Heat Transfer

2026-08-25

These are the notes for the obligatory Heat Transfer course on the 5th semester of the Bs.c. in Mechanical Engineering at Aarhus University taught by Pourya Forooghi and Mugisa Y. Kano.

## 1 Introduction to Heat Transfer

Heat transfer as a subject has already been treated in the [Partial Differential Equations Course](#). Further, a related subject, namely [Thermodynamics](#) has also been studied. However, whereas Thermodynamics mainly concerns itself with questions such as what the eventual equilibrium temperature will be or the total amount of heat transferred, heat transfer as a subject is more interested in questions such as the time it takes or the rate of heat transfer.

Heat Transfer is used in both power generation, heating/cooling/refrigeration, the chemical industry, indoor climate, thermal management and much more. Further, new technology means heat transfer will remain very useful into the future as data centers, electronics, and turbine blades all need to meet higher cooling standards to maintain performance.

Before we start, a quick note on terminology is useful. We distinguish between *Heat*, *Heat transfer rate*, and *Heat flux*. Heat,  $Q[\text{J}]$ , denotes the amount of heat transferred. The heat transfer rate,  $\dot{Q}[\frac{\text{J}}{\text{s}}]$ , denotes the heat  $Q$  transferred per unit time. And finally, Heat flux,  $\dot{q}[\frac{\text{W}}{\text{m}^2}]$ , is the heat transfer rate  $\dot{Q}$  per unit area.

### 1.1 Thermodynamics Recap

Here will be given a short Introduction to the most important concepts of Thermodynamics which we will be dealing with here. For a more in-depth review of the material see [my notes on the subject](#).

The change in internal energy  $u$  and enthalpy  $h = u + p/\rho$  with temperature is generally given as:

- Ideal Gasses:

$$\begin{aligned} du &= c_v dT \\ dh &= c_p dT. \end{aligned}$$

- Incompressible substances:

$$du = c dT \quad (c \approx c_p \approx c_v).$$

Also, the 1st law of thermodynamics (the energy balance) states that the net energy transfer by heat, work, and mass from a system equals the change in the total energy of said system. This also implies that the rate of energy transfer equals the rate of change in total energy.

## 2 Heat Transfer Mechanisms

We mainly work with three different heat transfer mechanisms. These are *Conduction*, *Convection*, and *Radiation*.

### 2.1 Conduction

Conduction is the transfer of energy from more energetic particles of a substance to the adjacent less energetic ones. This happens in both solids and fluids, but always through a material medium, so it cannot happen in a vacuum.

The rate of conduction  $\dot{Q}_{\text{cond}}$  is generally given on a form resembling:

$$\dot{Q}_{\text{cond}} \propto A \frac{\Delta T}{\Delta x}$$

where  $A$  is the conducting area,  $\Delta T$  is the change in temperature and  $\Delta x$  is the thickness of the cross-section. This can be qualified by including the *thermal conductivity*  $\left[ \frac{\text{W}}{\text{m K}} \right]$  which gives:

$$\dot{Q}_{\text{cond}} = -kA \frac{\Delta T}{\Delta x}.$$

As  $\Delta x \rightarrow 0$  this turns into *Fourier's law of conduction* (1D), given as:

$$\dot{q}_{\text{cond}} = -k \frac{dT}{dx} \quad (1)$$

or in general vector form as

$$\dot{\mathbf{q}} = -k \nabla T.$$

The thermal conductivity is a material property which varies with temperature. (Çengel and Ghajar 2015, fig. 1-23).

We also define the *thermal diffusivity*  $\alpha$  as

$$\alpha = \frac{k}{\rho c_p}.$$

Roughly speaking, the thermal diffusivity determines how fast heat diffuses in a certain solid material.

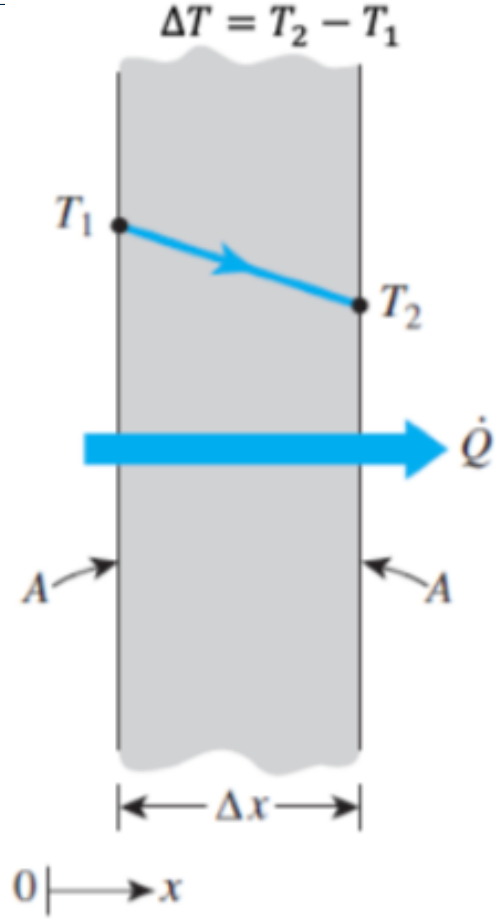
## 2.2 Convection

Convection is the transfer of energy as a result of bulk motion of fluid. This always happens between a solid surface and a flowing fluid. This can be both *natural (free) convection*, i.e. flow created by bouyancy force and *forced convection*, i.e. flow created by an external driving force. When a material undergoes convection it will generally also be undergoing convection to some degree.

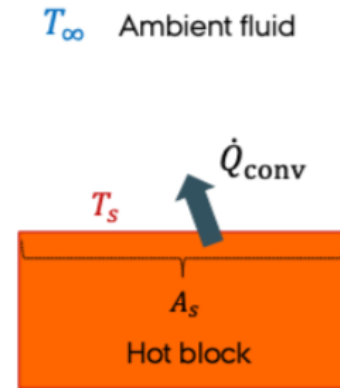
The convection rate  $\dot{Q}_{\text{conv}}$  is normally given in a form such as

$$\dot{Q}_{\text{conv}} = \alpha A_s (T_s - T_\infty)$$

where  $T_\infty$  is the far field temperature (or some other reference temperature),  $T_s$  is the surface temperature, and  $A_s$  is the



**Figure 2.1:** Cross section of a material displaying thermal conduction (Çengel and Ghajar 2015, fig. 1-23).



**Figure 2.2:** Cross section of a material displaying convection.

heat transfer surface area. By introducing the heat transfer coefficient  $h \left[ \frac{\text{W}}{\text{m}^2 \text{K}} \right]$  we arrive at Newton's law of cooling:

$$\dot{Q}_{\text{conv}} = hA_s(T_s - T_{\infty}). \quad (2)$$

It is important to remember the difference between the thermal conductivity  $k$  and the heat transfer coefficient  $h$ . Whereas the thermal conductivity is a material property the heat transfer coefficient is a property of the specific flow and must therefore be experimentally determined in all but the simplest cases.

## 2.3 Radiation

Thermal radiation is energy emitted by matter in form of electromagnetic waves. Here, there is no need for an intervening medium, so this may also happen in a vacuum. All bodies at a temperature above 0 K emit thermal radiation. In general, radiation is a volumetric phenomenon but for opaque solids it is usually considered a surface phenomenon. The heat transfer rate due to radiation,  $\dot{Q}_{\text{emit}}$ :

$$\dot{Q}_{\text{emit}} \propto A_s T_s^4.$$

For a black body (an ideal radiator) this is precisely:

$$\dot{Q}_{\text{emit}} = \sigma A_s T_s^4$$

where  $\sigma = 5.670 \cdot 10^{-8} \text{ W/m}^2/\text{K}^4$  is Boltzmann's constant. The above is also known as Boltzmann's law. For an arbitrary surface we instead have

$$\dot{Q}_{\text{emit}} = \epsilon \sigma A_s T_s^4$$

where  $0 \leq \epsilon \leq 1$  is the emissivity.

Emissivity  $\epsilon$  is a property of the radiating surface, which in reality varies with temperature but also depends on the wavelength of the electromagnetic wave. For simplicity, we often use its mean value.

The opposite case of emitted radiation is that of absorbed radiation given by

$$\dot{Q}_{\text{abs}} = \alpha \dot{Q}_{\text{incident}}$$

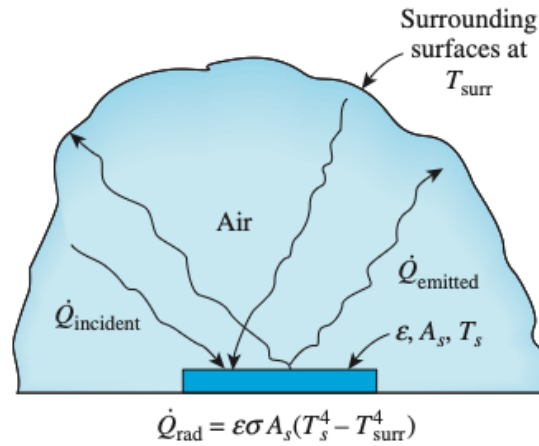
where  $0 \leq \alpha \leq 1$  is the absorptivity. Kirchoff's law of radiation states that at the same temperature and wavelength  $\epsilon = \alpha$ .

The net radiation heat transfer of a solid surface is

$$\dot{Q}_{\text{rad}} = \dot{Q}_{\text{emit}} - \dot{Q}_{\text{abs}}.$$

We have an important special case of the above for a surface entirely enclosed by a much larger surrounding surface as shown on [Figure 2.3](#), where

$$\dot{Q}_{\text{rad}} = \epsilon \sigma A_s (T_s^4 - T_{\text{surr}}^4).$$



**Figure 2.3:** Radiation between enclosed and enclosing body (Çengel and Ghajar 2015, fig. 1–40).

Lecture 2: Conduction, Pt. I

2026-09-01

### 3 Conduction

Heat transfer has both a *direction* and a *magnitude* and is therefore a vector quantity. The magnitude in a specified direction is proportional to the *temperature gradient*. Heat conduction in a medium is *steady*, when the temperature  $T = T(x, y, z)$  does not vary with time and *unsteady* or *transient* when  $T = T(x, y, z, t)$  varies with time. During transient heat transfer, the temperature normally varies with time as well as position. In the special case of variation with time but not with position, the temperature of the medium changes uniformly with time – such systems are termed *lumped systems*. Many small objects can be analyzed as being lumped systems.

In Section 2.1, heat conduction was defined as the transfer of thermal energy from more energetic particles of a medium to the adjacent less energetic ones. We generally employ a coordinate system such that heat transfer in the positive direction of a coordinate axis is positive and vice versa.

We previously introduced Fourier's law of heat conduction in Equation (1),

$$\dot{Q}_{\text{cond}} = -kA \frac{dT}{dx}$$

where  $k$  is the *thermal conductivity* and  $\frac{dT}{dx}$  is the temperature gradient. In general the thermal conductivity of a material  $k$  varies with time, however for most purposes an average value can be used for sufficiently accurate results.

Heat may also be generated throughout a medium. E.g. the temperature of a resistor rises volumetrically when a current is applied. Often it is also convenient to model the absorption of radiation such as solar energy or gamma rays as heat generation when those rays penetrate deep into a medium while being absorbed gradually such as for solar rays in a body of water. For opaque bodies almost all absorption occurs within the first few microns of the surface, so in these cases the solar energy can be treated as heat flux on the surface. As heat generation is volumetric, it is usually specified per unit volume and denoted by  $\dot{e}_{\text{gen}} = \left[ \frac{\text{W}}{\text{m}^3} \right]$ . The rate of heat generation may vary with time as well as position. When the variation w.r.t. the position is known, the total

rate of heat generation in a medium of volume  $\mathcal{V}$  can be determined from

$$\dot{E}_{\text{gen}} = \int_{\mathcal{V}} \dot{e}_{\text{gen}} d\mathcal{V}. \quad (3)$$

In the special case of uniform heat generation Equation (3) reduces to

$$\dot{E}_{\text{gen}} = \dot{e}_{\text{gen}} \mathcal{V}.$$

### 3.1 Heat Conduction Equation

It is useful to expand Equation (1) to more dimensions and include the internal heat generation. In rectangular coordinates it can be shown that Equation (1) can be written as

$$\frac{\partial}{\partial x} \left( k \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left( k \frac{\partial T}{\partial y} \right) + \frac{\partial}{\partial z} \left( k \frac{\partial T}{\partial z} \right) + \dot{e}_{\text{gen}} = \rho c \frac{\partial T}{\partial t}. \quad (4)$$

for density  $\rho$  and specific heat  $c$ . In cylindrical coordinates it can be written

$$\frac{1}{r} \frac{\partial}{\partial r} \left( kr \frac{\partial T}{\partial r} \right) + \frac{1}{r^2} \frac{\partial}{\partial \phi} \left( k \frac{\partial T}{\partial \phi} \right) + \frac{\partial}{\partial z} \left( k \frac{\partial T}{\partial z} \right) + \dot{e}_{\text{gen}} = \rho c \frac{\partial T}{\partial t}$$

and in spherical coordinates it is

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left( kr^2 \frac{\partial T}{\partial r} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial}{\partial \phi} \left( k \frac{\partial T}{\partial \phi} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial}{\partial \theta} \left( k \sin \theta \frac{\partial T}{\partial \theta} \right) + \dot{e}_{\text{gen}} = \rho c \frac{\partial T}{\partial t}.$$

Equation (4) is the general heat conduction equation in rectangular coordinates. In the case of constant thermal conductivity it reduces to:

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} + \frac{\dot{e}_{\text{gen}}}{k} = \nabla^2 T + \frac{\dot{e}_{\text{gen}}}{k} = \frac{1}{\alpha} \frac{\partial T}{\partial t} \quad (5)$$

where  $\alpha = k/\rho c$  is the *thermal diffusivity of the material*. Equation (5) is known as the Fourier-Biot Equation.

#### 3.1.1 Special Cases of the Heat Equation

For steady state systems the temperature does not vary with time  $\frac{\partial T}{\partial t} = 0$ , so for steady-state:

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} + \frac{\dot{e}_{\text{gen}}}{k} = \nabla^2 T + \frac{\dot{e}_{\text{gen}}}{k} = 0. \quad (6)$$

Equation (6) is known as the Poisson equation.

For transient systems with no heat generation Equation (5) reduces to:

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} = \nabla^2 T = \frac{1}{\alpha} \frac{\partial T}{\partial t}. \quad (7)$$

Equation (7) is also known as the diffusion equation.

For steady-state systems without heat generation we arrive at Section 3.1.1: Special Cases of the

Heat Equation also known as the Laplace equation:

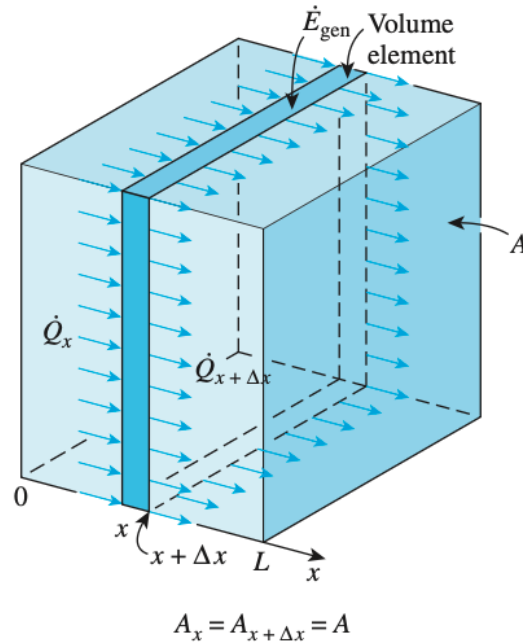
$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} = \nabla^2 T = 0.$$

### 3.2 One-Dimensional Steady Heat Conduction

We consider heat conduction through a large plane wall such as the wall of a house, the glass of a single pane window, or many other cases. Heat conduction in these conditions can be approximated as being *one-dimensional* since the conduction is dominant in one direction and negligible in the others.

#### 3.2.1 Plane Walls

We consider a thin element of thickness  $\Delta x$  in a large plane wall, as shown in Figure 3.1. We assume the density of the wall to be  $\rho$ , the specific heat to be  $c$  and the area of the wall normal to the direction of heat transfer to be  $A$ .

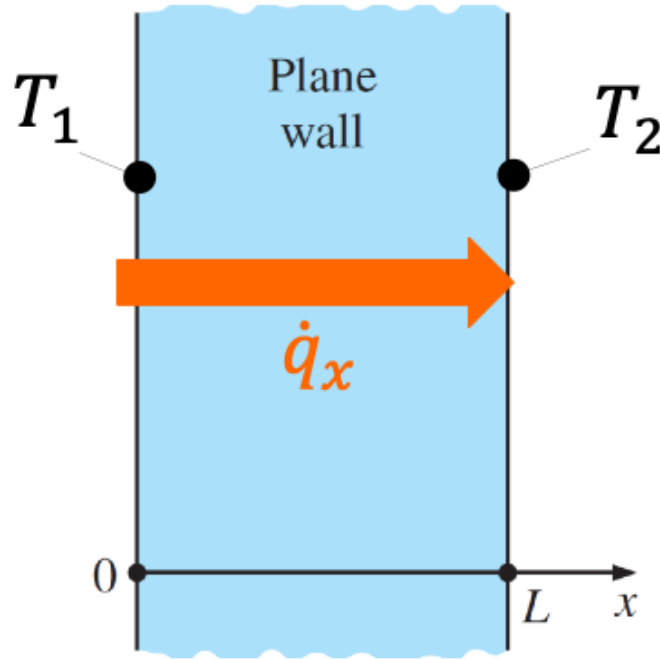


**Figure 3.1:** One-dimensional heat conduction through a volume element in a large plane wall.  
Reproduced from (Çengel and Ghajar 2015, fig 2–12).

Taking the direction normal to the surface of the wall, i.e. through the plane on Figure 3.1 to be the  $x$ -direction we obtain:

$$\begin{aligned}\frac{d^2 T}{dx^2} &= 0 \\ \frac{dT}{dx} &= C_1 \\ T &= C_1 x + C_2.\end{aligned}$$

On one side of the wall the temperature  $T(0) = T_1$  and expressing the thickness of the wall as  $L$  the boundary condition on the other side of the wall is  $T(L) = T_2$  as shown on Figure 3.2



**Figure 3.2:** Conduction through a 1D-wall.

As such

$$T(x) = T_1 + (T_2 - T_1) \frac{x}{L}.$$

This can then be used in Equation (1) as

$$\begin{aligned} \dot{q}_x &= -k \frac{dT}{dx} \\ \dot{q}_x(x) &= \text{const.} = k \frac{T_1 - T_2}{L} \\ \dot{Q} &= k \frac{T_1 - T_2}{L} A. \end{aligned}$$

### 3.2.2 Cylindrical Geometries

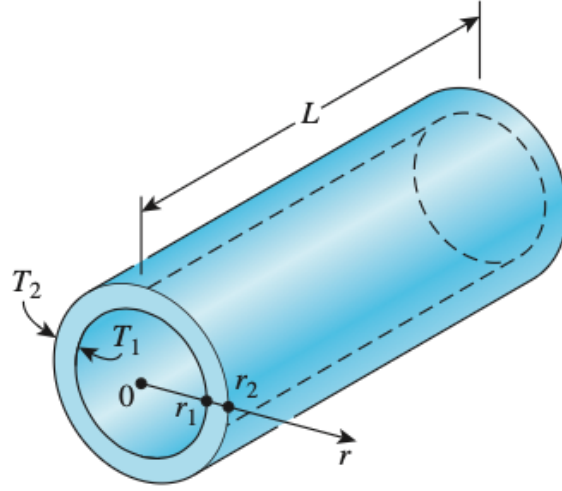
We consider a pipe of length  $L$ , inner and outer radiusses  $r_1$  and  $r_2$  and thermal conductivity  $k$  as shown on Figure 3.3.

The governing differential equation in this case is

$$\frac{d}{dr} \left( r \frac{dT}{dr} \right) = 0$$

with boundary conditions  $T(r_1) = T_1, T(r_2) = T_2$ . Integrating the differential equation with respect to  $r$  gives

$$\begin{aligned} r \frac{dT}{dr} &= C_1 \frac{dT}{dr} = \frac{C_1}{r} \\ T(r) &= C_1 \ln r + C_2. \end{aligned}$$



**Figure 3.3:** Schematic of a long cylinder with thickness  $\Delta r = r_2 - r_1$ . Reproduced from (Çengel and Ghajar 2015, fig. 2–50).

Applying the boundary conditions this becomes

$$T(r) = T_1 + (T_2 - T_1) \frac{\ln(r/r_1)}{\ln(r_2/r_1)}.$$

This can be used in Equation (1) as

$$\begin{aligned} \dot{q}_r(r) &= -k \frac{dT}{dr} = k \frac{T_1 - T_2}{\ln(r_2/r_1) r} \\ \dot{Q} &= 2\pi l k \frac{T_1 - T_2}{\ln(r_2/r_1)}. \end{aligned}$$

### 3.2.3 Internal Heat Generation

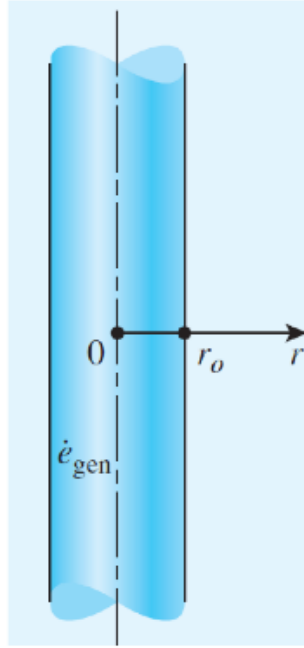
We now consider a long cylinder/wire which generates heat uniformly as shown on Figure 3.4. From steady state on cylindrical coordinates with uniform internal heat generatio we have

$$\begin{aligned} \frac{1}{r} \frac{d}{dr} \left( r \frac{dT}{dr} \right) + \frac{\dot{e}_{\text{gen}}}{k} &= 0 \\ \frac{d}{dr} \left( r \frac{dT}{dr} \right) &= -\frac{\dot{e}_{\text{gen}}}{k} r \\ r \frac{dT}{dr} &= -\frac{\dot{e}_{\text{gen}}}{k} \frac{r^2}{2} + C_1. \end{aligned}$$

Due to symmetry we have a boundary condition at  $r = 0$  as

$$\begin{aligned} \frac{dT}{dr}(r = 0) &= 0 \quad \implies \quad C_1 = 0 \\ \frac{dT}{dr} &= -\frac{\dot{e}_{\text{gen}}}{k} \frac{r}{2} \\ T &= -\frac{\dot{e}_{\text{gen}}}{k} \frac{r^2}{4} + C_2. \end{aligned}$$

Our second boundary condition is that the temperature at the edge is the surface temperature



**Figure 3.4:** Cylinder/wire conducting heat with uniform internal heat generation.

$T(r_0) = T_s$ . Hence

$$T = T_s + \frac{\dot{e}_{\text{gen}}}{4k} (r_0^2 - r^2)$$

$$T(r = 0) = T_s + \frac{\dot{e}_{\text{gen}}}{4k} r_0^2.$$

### 3.3 Boundary Conditions

A few different types of boundary conditions are normally employed

1. Specified Temperature (Dirichlet)
2. Specified heat flux (Neumann)
3. Insulated boundary – special case of Neumann with  $\dot{q}_0 = 0$
4. Thermal symmetry – Symmetric temperature profile
5. Convection (Robin / mixed) – Specified convection from surfaces

### 3.4 Thermal Resistance

Under steady conditions and surface temperatures the cumbersome differential equations can be avoided by instead introducing the concept of *thermal resistivity* analogous to electrical resistivity.

#### 3.4.1 Conduction Resistance

Analogous to electrical resistance:

$$I = \frac{V_1 - V_2}{R_e}$$

we now introduce thermal resistance

$$\dot{Q} = \frac{T_1 - T_2}{R}.$$

From the result for the conduction through a plane wall in Section 3.2.1: Plane Walls

$$\dot{Q} = k \frac{T_1 - T_2}{L} A$$

the thermal resistivity can by comparison with the definition be read to equal

$$R_{\text{cond, wall}} = \frac{L}{kA}.$$

From the result for the conduction through a cylinder from Section 3.2.2: Cylindrical Geometries

$$\dot{Q} = 2\pi l k \frac{T_1 - T_2}{\ln(r_2/r_1)}$$

we see that the thermal resistivity here is

$$R_{\text{cond, cyl}} = \frac{\ln(r_2/r_1)}{2\pi l k}.$$

### 3.4.2 Series Resistance Networks

The above conduction resistances can be combined. Say two walls are placed against each other. Following the above the thermal conduction through each wall is

$$\begin{aligned}\dot{Q}_2 &= \frac{T_1 - T_2}{R_{\text{wall},1}} \\ \dot{Q}_3 &= \frac{T_2 - T_3}{R_{\text{wall},2}}.\end{aligned}$$

We can then eliminate the intermediate temperature at the interface between the walls  $T_2$  as

$$\dot{Q}_{23} = \frac{T_1 - T_3}{R_{\text{wall},1} + R_{\text{wall},2}}.$$

We define the convective thermal resistance as

$$\dot{Q} = hA(T_\infty - T_s).$$

And as such for the case shown on Figure 3.5

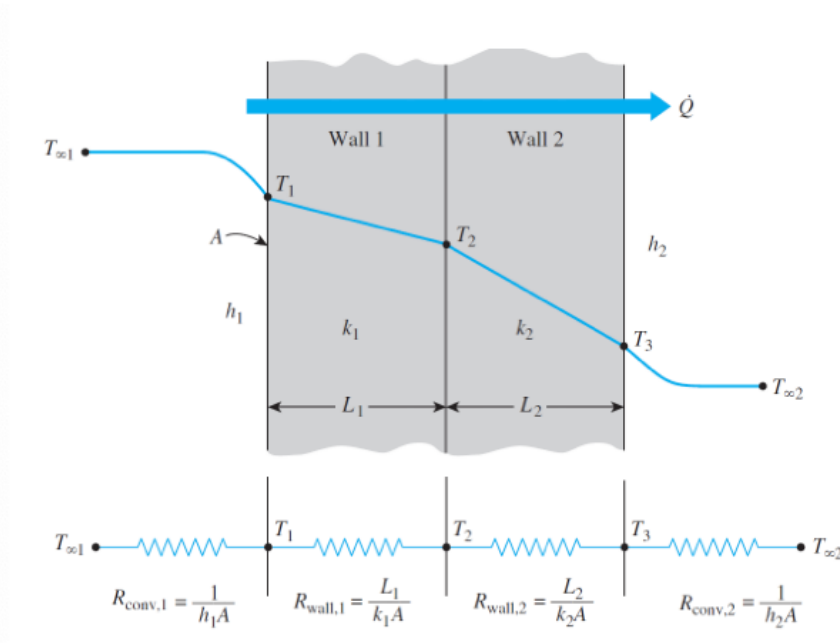
$$\begin{aligned}\dot{Q}_1 &= h_1 A (T_{\infty 1} - T_1) = \frac{T_{\infty 1} - T_1}{R_{\text{conv},1}} \\ \dot{Q}_4 &= h_2 A (T_3 - T_{\infty 2}) = \frac{T_3 - T_{\infty 2}}{R_{\text{conv},2}}.\end{aligned}$$

These can then all be combined to yield the total conduction through the wall

$$\dot{Q} = \frac{T_{\infty 1} - T_{\infty 2}}{R_{\text{conv},1} + R_{\text{conv},2} + R_{\text{wall},1} + R_{\text{wall},2}}.$$

### 3.4.3 Parallel and Generalized Resistance Networks

Analogous to electrical resistances thermal resistances can also be in parallel as shown on Figure 3.6. In such cases the total resistance,  $R_{\text{tot}}$  is calculated using the reciprocal of the reciprocal sums of

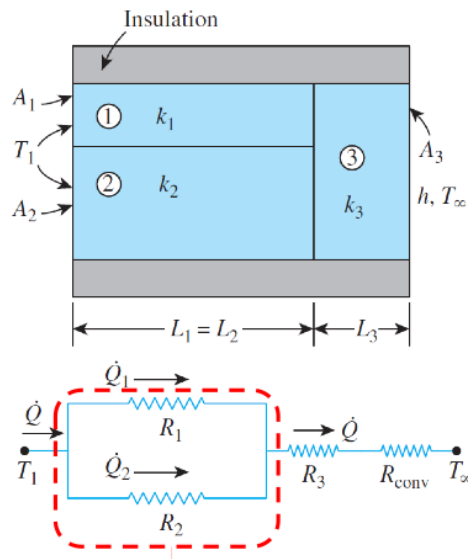


**Figure 3.5:** Thermal resistance in series.

the individual parallel resistances.

$$\frac{1}{R_{\text{tot}}} = \frac{1}{R_1} + \frac{1}{R_2}.$$

Parallel thermal resistances only occur in 2D or 3D whilst thermal resistance theory is based on 1D solutions so this is only an approximation.



**Figure 3.6:** Parallel thermal resistance

### 3.4.4 Thermal Contact Resistance

In the above, we assumed “perfect contact” at the interface between layers, and thus no temperature drop at the interface. In reality no such interface exists and as a result any interface will contain imperfections and air gaps. Thus, an interface offers some resistance to heat transfer which is characterised by a *thermal contact resistance*  $R_c$  for a unit interface area. This can be expressed in

a manner analogous to Newton's law of cooling

$$\dot{Q} = h_c A \Delta T_{\text{interface}}$$

where  $A$  is the apparent interface area,  $\Delta T_{\text{interface}}$  is the effective temperature difference at the interface and  $h_c$  is the *thermal contact conductance* defined as

$$h_c = \frac{\dot{Q}/A}{\Delta T_{\text{interface}}}$$

which is the inverse of thermal contact resistance  $R_c$

$$R_c = \frac{1}{h_c} = \frac{\Delta T_{\text{interface}}}{\dot{Q}/A}.$$

It is also worth noting that the heat transfer rate is the quotient of the temperature difference over the interface  $\Delta T_{\text{interface}}$  and the created thermal resistance  $R_c/A$  as

$$\dot{Q} = \frac{\Delta T_{\text{interface}}}{R_c/A}.$$

Lecture 3: Conduction, Pt. II

2026-09-8

### 3.5 Steady Heat Conduction in 2D & 3D

#### 3.5.1 Analytical and Numerical Solutions

For multidimensional steady conduction, the governing equation is the steady heat conduction equation,

$$\nabla^2 T + \frac{\dot{e}_{\text{gen}}}{k} = 0..$$

Analytical solutions exist for relatively simple geometries and boundary conditions, but rapidly become cumbersome. Practical multidimensional problems are therefore commonly solved numerically or by use of existing analytical solutions. These existing analytical solution are often given in term of a shape factor  $S$ [m].

#### 3.5.2 Conduction Shape Factors

The shape factor is defined such, that the stationary heat flow:

$$\begin{aligned} q_x'' &= -k \frac{\partial T}{\partial x} \\ q_y'' &= -k \frac{\partial T}{\partial y} \end{aligned}$$

can be written on a simple form:

$$Q = Sk\Delta T_{1-2}. \quad (8)$$

Hence, the shape factors can only be used when considering conduction between two surfaces with constant temperature. The conduction shape factors depend only on the *geometry* of the system. These conduction shape factors  $S$  have been determined for a number of standard configurations and a small excerpt of these are shown in (Çengel and Ghajar 2015, Table 3–7). These are also

only applicable when the mode of heat transfer is conduction and they are therefore not useful for liquids or gasses. By observation of Equation (8) it is evident that the conduction shape factor  $S$  is related to thermal resistance  $R$  by

$$R = \frac{1}{kS} \quad \Longleftrightarrow \quad S = \frac{1}{kR}.$$

### 3.6 Extended and Finned Surfaces

As mentioned previously, the rate of heat transfer from a surface at temperature  $T_s$  to the surrounding medium at temperature  $T_\infty$  is per Newton's law of cooling, Equation (2):

$$\dot{Q}_{\text{conv}} = hA_s(T_s - T_\infty).$$

Hence, the heat transfer rate can be increased by either increasing the temperature difference, the convection heat transfer coefficient  $h$  or the surface area  $A_s$ . The temperature difference is often fixed by design constraints and increasing  $h$  may require installing a pump or fan or replacing an existing one with a larger one, but this may also not always be practical. The other alternative is increasing surface area by attaching *extended surfaces* called *fins* made of highly conductive material to the surface. The addition of fins often increase the heat transfer from a surface severalfold.

#### 3.6.1 Fin Equation

From an energy balance on a fin, using Fourier's law of heat conduction it can be shown that the differential equation governing heat transfer in fins is:

$$\frac{d}{dx} \left( kA_c \frac{dT}{dx} \right) = hp(T - T_\infty).$$

In general, the cross sectional area  $A_c$  and the perimeter  $p$  of a fin vary with  $x$  which makes the above differential equation hard to solve. For *constant cross section* and *constant thermal conductivity* the above reduces to:

$$\frac{d^2T}{dx^2} = \frac{hp}{kA_c}(T - T_\infty).$$

This can also be expressed as

$$\frac{d^2\theta}{dx^2} = m^2\theta \tag{9}$$

where

$$m^2 = \frac{hp}{kA_c}$$

and  $\theta = T - T_\infty$  is the *temperature excess*. As  $m$  in Equation (9) is assumed constant the differential equation can be solved yielding

$$\theta(x) = C_1 e^{mx} + C_2 e^{-mx}$$

where  $C_1$  and  $C_2$  are arbitrary constants whose values depend on the boundary conditions at the base and at the tip of the fin.

The temperature of the plate to which the fin is attached is normally known in advance. Therefore,

at the fin base we have a Dirichlet BC, expressed as

$$\theta(0) = \theta_b = T_b - T_\infty.$$

At the fin tip, however, we have several possibilities, which are explained in Section 3.6.2: Boundary Conditions for Fins.

### 3.6.2 Boundary Conditions for Fins

**Infinitely Long Fins** For a sufficiently long fin of uniform cross section, the temperature of the fin at the fin tip  $T(L)$  approaches the environment temperature  $T_\infty$  and thus  $\theta = T(L) - T_\infty \rightarrow 0$ . Therefore, for an infinitely long fin we have the boundary condition at the fin tip:

$$\theta(L) = T(L) - T_\infty = 0, \quad L \rightarrow \infty.$$

In this case our general solution after applying BC's become:

$$\theta(x) = \theta_b e^{-mx} = \theta_b e^{-x \sqrt{\frac{hp}{kA_c}}}.$$

This corresponds to the temperature decreasing exponentially from  $T_b$  to  $T_\infty$ . The steady rate of heat transfer for an infinitely long fin can be determined from Fourier's law of heat conduction

$$\dot{Q}_{\text{long fin}} = -kA_c \left. \frac{dT}{dx} \right|_{x=0} = \sqrt{hpkA_c} \theta_b$$

where  $p$  is the perimeter,  $A_c$  is the cross-sectional area of the fin, and  $x$  is the distance from the fin base.

**Negligible Heat Loss from the Fin Tip** Few fins are actually sufficiently long that their temperature approaches the surrounding temperature at the tip. A more realistic expectation is for heat transfer from the fin tip to be negligible since the heat transfer is proportional to surface area and little of the total surface area is in the tip. In such cases, the fin tip can be assumed adiabatic yielding the following boundary condition at the fin tip:

$$\left. \frac{d\theta}{dx} \right|_{x=L} = 0.$$

After applying both BCs to the solution to the differential equation we get the following expression for the temperature excess along the fin:

$$\theta(x) = \theta_b \frac{\cosh m(L-x)}{\cosh mL}$$

and the rate of heat transfer can, again, be determined from Fourier's law of heat conduction:

$$\dot{Q}_{\text{adiabatic tip}} = -kA_c \left. \frac{dT}{dx} \right|_{x=0} = \sqrt{hpkA_c} \theta_b \tanh mL.$$

**Specified Temperature** Instead of assuming the fin tip does not transfer heat or that the fin is sufficiently long for the fin tip to reach the surrounding temperature, it is often more accurate

to specify a temperature at the fin tip which is assumed to be fixed. This temperature is often a bit higher than room temperature but the temperature excess at the tip under normal operating conditions must often be determined exponentially. In this case, the boundary condition for the fin tip is

$$\theta(L) = \theta_L = T_L - T_\infty.$$

In this case the temperature distribution along the wing is described by

$$\theta(x) = \theta_b \frac{\left(\frac{\theta_L}{\theta_b}\right) \sinh mx + \sinh m(L-x)}{\sinh mL}.$$

From Fourier's law of heat conduction, the rate of heat transfer from the fin in this case is

$$\dot{Q}_{\text{specified temperature}} = -kA_c \left. \frac{dT}{dx} \right|_{x=0} = \sqrt{hpkA_c} \theta_b \frac{\cosh mL - \frac{\theta_L}{\theta_b}}{\sinh mL}.$$

**Convection from Fin Tip** In actuality, fin tips are exposed to the surroundings, and thus the proper boundary conditions for the fin tip is convection that also may include the effects of radiation. In this case, the boundary condition at the fin tip can be obtained from an energy balance at the fin tip ( $\dot{Q}_{\text{cond}} = \dot{Q}_{\text{conv}}$ ). This yields the boundary condition:

$$-kA_c \left. \frac{dT}{dx} \right|_{x=L} = hA_c \theta_L.$$

In this case, the temperature profile across the fin is

$$\theta(x) = \theta_b \frac{\cosh m(L-x) + \left(\frac{h}{mk}\right) \sinh m(L-x)}{\cosh mL + \left(\frac{h}{mk}\right) \sinh mL}.$$

The rate of heat transfer here is

$$\dot{Q}_{\text{convection}} = -kA_c \left. \frac{dT}{dx} \right|_{x=0} = \sqrt{hpkA_c} \theta_b \frac{\sinh mL + \frac{h}{mk} \cosh mL}{\cosh mL + \frac{h}{mk} \sinh mL}.$$

### 3.6.3 Fin Efficiency

We now consider the surface of a *plane wall* at temperature  $T_b$  exposed to a medium at temperature  $T_\infty$ . Heat is lost from the surface to the surrounding medium by convection with a heat transfer coefficient of  $h$ . Disregarding radiation, heat transfer from a surface area  $A_s$  is  $\dot{Q} = hA_s(T_b - T_\infty)$ .

Now we attach a fin of constant cross sectional-area  $A_c$  and length  $L$  to the surface with perfect contact. Now, heat is transferred from the surface to the fin by conduction and from the fin to the medium by convection with the same heat transfer coefficient  $h$ . The temperature of the fin is  $T_b$  at the base and gradually decreases toward the tip. Convection from the fin causes the temperature to drop somewhat from the midsection toward the outer surface. However, the cross-sectional area of the fin is usually very small and thus the temperature at any cross section can be assumed uniform.

In the limiting case of zero thermal resistance or infinite thermal conductivity, the temperature of

the fin is uniform at the base at  $T_b$ . The heat transfer from the fin is maximum in this case and can be expressed as

$$\dot{Q}_{\text{fin, max}} = hA_{\text{fin}}(T_b - T_{\infty}).$$

In reality, the temperature of the fin drops along the fin and thus the heat transfer from the fin is less because of the decreasing temperature excess  $T(x) - T_{\infty}$ . To account for this, we define fin efficiency as

$$\eta_{\text{fin}} = \frac{\dot{Q}_{\text{fin}}}{\dot{Q}_{\text{fin, max}}} = \frac{\text{actual heat transfer rate from the fin}}{\text{Heat transfer rate from the fin if the entire fin was at base temperature}}.$$

Hence, the total heat transfer from a finned surface is

$$\dot{Q}_{\text{fin}} = \eta_{\text{fin}} \underbrace{A_{\text{fin}} h \theta_b}_{\dot{Q}_{\text{fin, max}}}.$$

And the total heat transfer is

$$\begin{aligned} Q_{\text{total, fin}} &= \dot{Q}_{\text{fin}} + \dot{Q}_{\text{unfin}} \\ &= \eta_{\text{fin}} A_{\text{fin}} h \theta_b + A_{\text{unfin}} h \theta_b \\ &= (\eta_{\text{fin}} A_{\text{fin}} + A_{\text{unfin}}) h \theta_b. \end{aligned}$$

The fin efficiency in the cases of *very long fins* and *adiabatic tips* can be expressed as:

$$\begin{aligned} \eta_{\text{ling fin}} &= \frac{1}{mL} \\ \eta_{\text{adiabatic tip}} &= \frac{\tanh mL}{mL}. \end{aligned}$$

### 3.6.4 Fin Effectiveness

There is no assurance that adding fins to a surface will *enhance* heat transfer. The performance of the fins is judged on the basis of the enhancement in heat transfer relative to the *no-fin* case. The performance of fins is expressed in terms of the fin effectiveness  $\epsilon_{\text{fin}}$ :

$$\epsilon_{\text{fin}} = \frac{\dot{Q}_{\text{fin}}}{\dot{Q}_{\text{no fin}}} = \frac{\dot{Q}_{\text{fin}}}{hA_b\theta_b}.$$

Hence,  $A_b$  is the cross sectional area of the fin at the base and  $\dot{Q}_{\text{no fin}}$  represents the rate of heat transfer from this area if no fin were attached. An effectiveness of  $\epsilon_{\text{fin}} = 1$  indicates that the addition of fins to the surface does not affect heat transfer at all.

The fin efficiency and fin effectiveness are different quantities but related to each other by:

$$\epsilon_{\text{fin}} = \frac{A_{\text{fin}}}{A_b} \eta_{\text{fin}}.$$

The overall effectiveness,  $\epsilon_{\text{fin, overall}}$  is the effectiveness of the addition of fins to the entire area:

$$\epsilon_{\text{fin, overall}} = \frac{\dot{Q}_{\text{total, fin}}}{\dot{Q}_{\text{total, no fin}}} = \frac{\eta_{\text{fin}} A_{\text{fin}} + A_{\text{unfin}}}{A_b + A_{\text{unfin}}}.$$

## Lecture 4: Conduction, Pt. III

2026-09-15

### 3.7 Transient Heat Conduction

In transient heat conduction, the temperature field changes with time,

$$T = T(x, y, z, t).$$

The governing heat equation therefore retains the transient term

$$\rho c \frac{\partial T}{\partial t}.$$

#### 3.7.1 Lumped System Analysis

In heat transfer analysis, some bodies are observed to behave like a “lump”, i.e. whose interior temperature remains uniform at any point during a heat transfer process. The temperature of such bodies can be taken to be a function of time only,  $T(t)$ . Analysis utilizing this idealization is known as *lumped system analysis*. This is often applicable for small bodies.

We consider a body of arbitrary shape of mass  $m$ , volume  $\mathcal{V}$ , surface area  $A_s$ , density  $\rho$ , and specific heat  $c_p$  initially at uniform temperature  $T_i$ . At time  $t = 0$ , the body is placed into a medium at temperature  $T_\infty$ , and heat transfer takes place between the body and the medium with heat transfer coefficient  $h$ . We assume the system to be lumped, such that the temperature within the body remains uniform,  $T = T(t)$ . During a differential time element  $dt$ , the temperature of the body rises by a differential amount  $dT$ . An energy balance of the solid for the time interval  $dt$  can be expressed as

$$hA_s(T_\infty - T) dt = mc_p dT.$$

Noting that  $m = \rho\mathcal{V}$  and  $dT = d(T - T_\infty)$  since  $T_\infty = \text{constant}$  the above can be rearranged as

$$\frac{d(T - T_\infty)}{T - T_\infty} = -\frac{hA_s}{\rho\mathcal{V}c_p} dt.$$

integrating from  $t = 0$ , at which  $T = T_i$  to a time  $t$  at which  $T = T(t)$  gives

$$\ln \frac{T(t) - T_\infty}{T_i - T_\infty} = -\frac{hA_s}{\rho\mathcal{V}c_p} t.$$

Rearranging we obtain

$$T - T_\infty = (T_i - T_\infty)e^{-bt} \quad (10)$$

where

$$b = \frac{hA_s}{\rho\mathcal{V}c_p} \quad \frac{1}{s}$$

is a positive quantity. The reciprocal of  $b$ , is termed  $\tau$  and is called the time constant defined as:

$$\tau = \frac{1}{b} = \frac{\rho c_p \mathcal{V}}{hA_s} \quad \implies \quad \theta = e^{-\frac{t}{\tau}}.$$

**Heat Transfer Rate and Total Heat Transfer** Once the temperature  $T(t)$  at time  $t$  is found from Equation (10), the rate of convection heat transfer between the body and its environment at

that time can be determined from Newton's law of cooling as

$$\dot{Q}(t) = hA_s(T(t) - T_\infty).$$

The *total amount* of heat transfer between the body and the surrounding medium over the time interval  $t = 0$  to  $t$  is simply the change in the energy content of the body:

$$Q = mc_p(T(t) - T_i).$$

The amount of heat transfer reaches its upper limit when the body reaches the surrounding temperature, therefore the *maximum* heat transfer between the body and its surroundings is

$$Q_{\max} = mc_p(T_\infty - T_i).$$

### 3.7.2 Biot Number and Lumped-System Criterion

Now, we must determine when lumped analysis is appropriate to use. The first step in establishing such a criterion is to define a characteristic length as:

$$L_c = \frac{\mathcal{V}}{A_s}$$

and a dimensionless *Biot number* Bi as

$$\text{Bi} = \frac{hL_c}{k}. \quad (11)$$

The characteristic length  $L_c$  to be used in the Biot number for simple geometries in which heat transfer is one-dimensional, such as a large plane wall of thickness  $2L$ , a long cylinder of radius  $r_o$ , and a sphere of radius  $r_o$  becomes  $L$ ,  $r_o/2$ , and  $r_o/3$ , respectively. The Biot number is the ratio of internal resistance of a body to heat conduction to its external resistance to heat convection. Therefore a small Biot number represents a small resistance to conduction and thus small temperature gradients within the body.

Lumped system analysis assumes a uniform temperature distribution throughout the body. Thus lumped system analysis is exact when  $\text{Bi} = 0$  and approximate when  $\text{Bi} > 0$ . It is generally accepted that lumped system analysis is applicable if

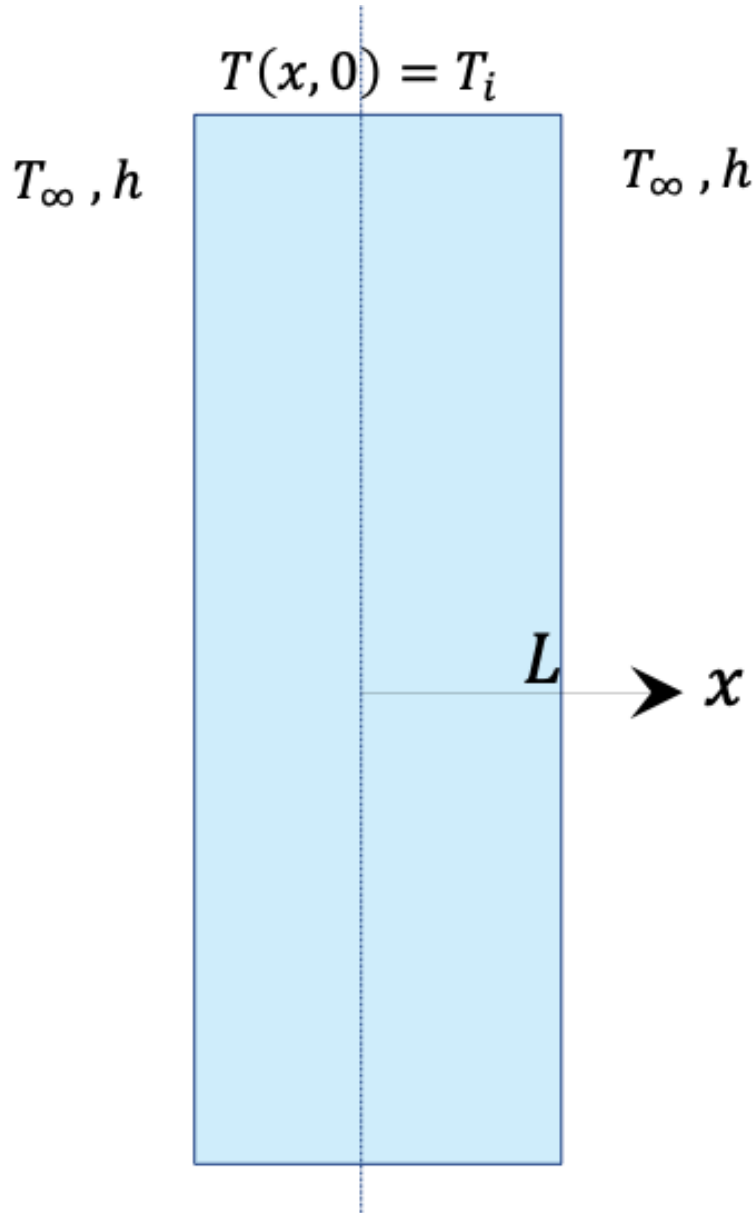
$$\text{Bi} \leq 0.1.$$

### 3.7.3 One-Dimensional Transient Heat Conduction

We now move on to transient heat conduction in non-lumped bodies one-dimensional bodies. We now consider the variation of temperature with *time* and *position* in one-dimensional problems such as those associated with a large plane wall, a long cylinder, and a sphere.

**Transient Conduction in a Plane Wall** We consider a wall of total thickness  $2L$ . The  $x$ -coordinate is measured from the centre plane, so  $x = 0$  is the middle and  $x = L$  is the exposed face. The same fluid is on both sides.

The left side at  $x = 0$  is insulated (i.e. no heat leaves) and the right side  $x = L$  loses heat to the environment through convection. The initial state is everything starting at the same temperature.



**Figure 3.7:** Transient Conduction in a Plane Wall.

Our governing differential equation is

$$\frac{\partial^2 T}{\partial x^2} = \frac{1}{\alpha} \frac{\partial T}{\partial t}, \quad \alpha = \frac{k}{\rho c_p}.$$

**Dimensionless Temperature, Biot Number and Fourier Number** Solving the above equation with the given BC's and IC would give  $T$  as a function of  $x$ ,  $t$ ,  $T_i$ ,  $T_{\text{inf}}$ ,  $k$ ,  $\alpha$ ,  $h$  and  $L$ . Instead, we can also opt to use dimensional analysis to make the problem dimension-less. First, we define the dimensionless temperature:

$$\theta = \frac{T - T_\infty}{T_i - T_\infty}$$

this is the temperature excess.

We also define the Fourier number, Fo:

$$\text{Fo} = \frac{\alpha t}{L^2} = \frac{t}{L^2/\alpha}.$$

The Fourier number can be seen as the ratio of elapsed time to the time required for heat to diffuse through the body. So a small Fo means the process just started and a large Fo means diffusion has had a long time to act.

Using the Fourier and previously introduced Biot numbers the temperature excess of the wall can be written as:

$$\theta(x, t) = \frac{T - T_\infty}{T_i - T_\infty} = \underbrace{f_1(\text{FoBi})}_{\text{Time part}} \cdot \underbrace{f_2\left(\frac{x}{L}, \text{Bi}\right)}_{\text{Position part}}.$$

**One-Term Approximation** The exact solution to the above is an infinite series. For  $\text{Fo} > 0.2$  is the first term of the series, however, normally enough. Then:

$$\begin{aligned} f_1(\text{Fo}, \text{Bi}) &\approx A_1 e^{-\lambda_1^2 \text{Fo}} \\ f_2\left(\frac{x}{L}, \text{Bi}\right) &\approx \cos\left(\lambda_1 \frac{x}{L}\right). \end{aligned}$$

Meaning

$$\theta = \frac{T - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \text{Fo}} \cdot \cos\left(\lambda_1 \frac{x}{L}\right)$$

where,  $A_1$  and  $\lambda_1$  are functions of Bi which are tabulated in (Çengel and Ghajar 2015, Table 4–2).

At the center of the wall,  $x = 0$  the temperature  $T_0$  is

$$\frac{T_0 - T_\infty}{T_i - T_\infty} = A_1 e^{-\lambda_1^2 \text{Fo}}.$$

When the center temperature has been determined, the spatial part can be used:

$$T(x, t) - T_\infty = (T_0(t) - T_\infty) \cos\left(\lambda_1 \frac{x}{L}\right).$$

So first, one must use the initial temperature to determine how much the center has heated/cooled at time  $t$  and then one uses the spatial profile to find the temperature near the surface or at any other point in the wall.

Instead of calculating the values of  $A_1 e^{-\lambda_1^2 \text{Fo}}$  and  $\cos(\lambda_1 x/L)$  as shown above one can instead opt to read the values from so-called Heisler charts. These are just lookup-plots for the above equations.

**Total Heat Transfer** In lumped systems the entire body has uniform temperature so the total heat transfer is easily found, however know the temperature varies with position  $T = T(x, t)$ . In this case, the total energy change has to be integrated instead:

$$Q(t) = \int_{\mathcal{V}} \rho c (T(x, t) - T_i) d\mathcal{V}.$$

After infinite time, the entire body reaches  $T_\infty$  and therefore:

$$Q_{\max} = mc(T_\infty - T_i)$$

so  $Q/Q_{\max}$  tells the fraction of total energy transfer which has already happened. For the plane wall one-term solution:

$$\frac{Q}{Q_{\max}} = 1 - \theta_0 \frac{\sin \lambda_1}{\lambda_1}.$$

## References

Çengel, Yunus A. and Afshin J. Ghajar (2015). *Heat and Mass Transfer: Fundamentals & Applications*. 5th edition. New York, NY: McGraw-Hill Education. 968 pp. ISBN: 978-0-07-339818-1.